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✓ Quick start with Hugging Face



[Run in Google Colab](#)



[Run in Colab Enterprise](#)



[View on GitHub](#)



[View on Hugging Face](#)

This notebook provides a basic demo of using MedGemma, a collection of Gemma 3 variants that are trained for performance on medical text and image comprehension. MedGemma is intended to accelerate building healthcare-based AI applications.

Learn more about the model at the [HAI-DEF developer site](#).

✓ Setup

To complete this tutorial, you'll need to have a runtime with [sufficient resources](#) to run the MedGemma model.

You can try out MedGemma 4B for free in Google Colab using a T4 GPU:

1. In the upper-right of the Colab window, select ▼ **(Additional connection options)**.
2. Select **Change runtime type**.
3. Under **Hardware accelerator**, select **T4 GPU**.

Note: To run the demo with MedGemma 27B in Google Colab, you will need a runtime with an A100 GPU.

Get access to MedGemma

Before you get started, make sure that you have access to MedGemma models on Hugging Face:

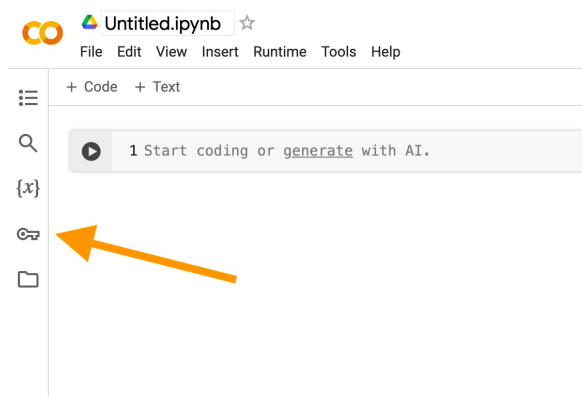
1. If you don't already have a Hugging Face account, you can create one for free by clicking [here](#).
2. Head over to the [MedGemma model page](#) and accept the usage conditions.

✓ Authenticate with Hugging Face

Generate a Hugging Face `read` access token by going to [settings](#).

If you are using Google Colab, add your access token to the Colab Secrets manager to securely store it. If not, proceed to run the cell below to authenticate with Hugging Face.

1. Open your Google Colab notebook and click on the  Secrets tab in the left



panel.

2. Create a new secret with the name `HF_TOKEN`.
3. Copy/paste your token key into the Value input box of `HF_TOKEN`.
4. Toggle the button on the left to allow notebook access to the secret.

```
import os
import sys

google_colab = "google.colab" in sys.modules and not os.environ.get("

if google_colab:
    # Use secret if running in Google Colab
    from google.colab import userdata
    os.environ["HF_TOKEN"] = userdata.get("HF_TOKEN")
else:
    # Store Hugging Face data under `/content` if running in Colab En
    if os.environ.get("VERTEX_PRODUCT") == "COLAB_ENTERPRISE":
```

```

os.environ["HF_HOME"] = "/content/hf"
# Authenticate with Hugging Face
from huggingface_hub import get_token
if get_token() is None:
    from huggingface_hub import notebook_login
    notebook_login()

```

✓ Install dependencies

```
! pip install --upgrade --quiet accelerate bitsandbytes transformers
```

✓ Load model from Hugging Face Hub

```

from transformers import BitsAndBytesConfig
import torch

model_variant = "medgemma-1.5-4b-it"
model_id = f"google/{model_variant}"

# @markdown ---
# @markdown To reduce memory usage, quantize the model in 4-bit.
use_quantization = True # @param {type: boolean}

# @markdown ---
# @markdown MedGemma 1.5 4B and MedGemma 27B variants support thinking. If enabled, a system instruction will be added to prompt the model to output its reasoning between two special tokens <unused94> and <unused95> followed by the final response.
is_thinking = False # @param {type: boolean}
THINKING_VARIANTS = [
    "medgemma-1.5-4b-it",
    "medgemma-27b-it",
    "medgemma-27b-text-it",
]
if is_thinking and model_variant in THINKING_VARIANTS:
    is_thinking = False
    print(
        "Note: Thinking is enabled. Set `is_thinking` to `False` to disable."
    )

# If running a 27B variant in Google Colab, check for GPU memory requirements
if "27b" in model_variant and google.colab.is_gpu_available():
    if not ("A100" in torch.cuda.get_device_capability()):
        raise ValueError(
            "Runtime has insufficient GPU memory. Please select an A100 GPU."
        )

```

model_variant

medgemma-1.5-4b-it ▼

To reduce memory usage, [quantize](#) the model in 4-bit. **Note:** 4-bit quantization is required to run MedGemma 27B on an A100 GPU in Google Colab. The performance of quantized versions has not been evaluated.

☐ use_quantization

MedGemma 1.5 4B and MedGemma 27B variants support thinking. If enabled, a system instruction will be added to prompt the model to output its reasoning between two special tokens `<unused94>` and `<unused95>` followed by the final response. **Note:** This notebook shows how to enable thinking for any multimodal or text-only task but it may not benefit some use cases. Thinking may potentially improve

```

model_kwargs = dict(
    dtype=torch.bfloat16,
    device_map="auto",
)

if use_quantization:
    model_kwargs["quantization_"]

```

results on challenging text-only tasks at the cost of generating more tokens. Thoroughly validate the model's performance and assess the inference time and cost when enabling thinking.

☐ is_thinking

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The following sections contain standalone examples demonstrating how to use the model both directly and with the [pipeline](#) API. The [pipeline](#) API provides a simple way to use the model for inference while abstracting away complex details, while directly using the model gives you complete control over the inference process, including preprocessing and postprocessing. In practice, you should select the method that is best suited for your use case.

Here, you will load the model directly and with the [pipeline](#) API for use in the next sections. Note that the multimodal variants and the 27B text-only variant are loaded with their respective tasks and classes.

Load model with the [pipeline](#) API

```

from transformers import pipeline

if "text" in model_variant:
    pipe = pipeline("text-generation", model=model_id, model_kwargs=model_kwargs)
else:
    pipe = pipeline("image-text-to-text", model=model_id, model_kwargs=model_kwargs)

```

Load model directly

```

if "text" in model_variant:
    from transformers import AutoModelForCausalLM, AutoTokenizer
    model = AutoModelForCausalLM.from_pretrained(model_id, **model_kwargs)
    tokenizer = AutoTokenizer.from_pretrained(model_id)
else:
    from transformers import AutoModelForImageTextToText, AutoProcessor
    model = AutoModelForImageTextToText.from_pretrained(model_id, **model_kwargs)
    processor = AutoProcessor.from_pretrained(model_id)

```

✓ Run inference on images and text

This section demonstrates running inference on image-based tasks using multimodal variants.

Note: Proceed to [Run inference on text only](#) if you have selected the 27B text-only variant.

```
if "text" in model_variant:
    raise ValueError(
        "You are using a text-only variant which does not support mul-
        tiple inputs. Please proceed to the 'Run inference on text only' s
    )
```

Specify image and text inputs

```
import os
from PIL import Image
from IPython.display import Image

prompt = "Describe this X-ray"

# Image attribution: Stillwater:
image_url = "https://upload.wikimedia.org/wikipedia/commons/4/4c/Stillwater.jpg"
! wget -nc -q {image_url}
image_filename = os.path.basename(image_url)
image = Image.open(image_filename)
```

prompt

Describe this X-ray

image_url

https://upload.wikimedia.org/wikipedia/commons/4/4c/Stillwater.jpg

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Format conversation

```
role_instruction = "You are an expert radiologist."
if is_thinking:
    system_instruction = f"SYSTEM INSTRUCTION: think silently if needed"
    max_new_tokens = 1300
else:
    system_instruction = role_instruction
    max_new_tokens = 300

messages = [
    {
        "role": "system",
        "content": [{"type": "text", "text": system_instruction}]
    },
    {
        "role": "user",
```

```

    role: "user",
    "content": [
      {"type": "text", "text": prompt},
      {"type": "image", "image": image}
    ]
  }
]

```

Run model with the `pipeline` API

```

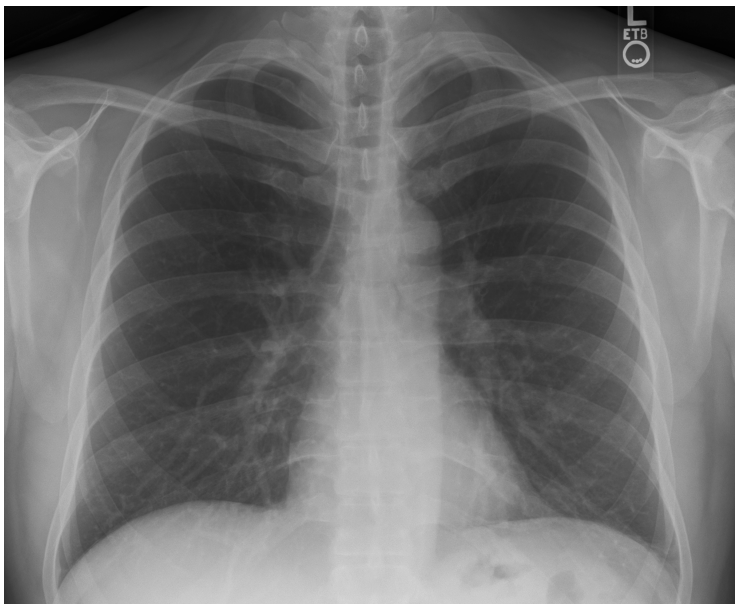
output = pipe(text=messages, max_new_tokens=max_new_tokens, do_sample=False)
response = output[0]["generated_text"][-1]["content"]

display(Markdown(f"---\n\n**[ User ]**\n\n{prompt}"))
display(IPImage(filename=image_filename, height=300))
if is_thinking:
    thought, response = response.split("<unused95>")
    thought = thought.replace("<unused94>thought\n", "")
    display(Markdown(f"---\n\n**[ MedGemma thinking trace ]**\n\n{thought}"))
display(Markdown(f"---\n\n**[ MedGemma ]**\n\n{response}\n\n---"))

```

[User]

Describe this X-ray



[MedGemma]

Based on the provided image, here's a description of the X-ray:

Overall Impression:

The image appears to be a standard PA (posterior-anterior) chest X-ray. The lungs are well-expanded, and there are no obvious signs of acute pathology.

Specific Findings:

- **Heart Size:** The heart size appears within normal limits.
- **Mediastinum:** The mediastinum (the space between the lungs containing the

heart, great vessels, and trachea) appears unremarkable.

- **Lungs:** The lung fields are clear, with no evidence of consolidation, effusions, or masses.
- **Bones:** The ribs, clavicles, and spine appear intact.
- **Soft Tissues:** The soft tissues of the chest wall are unremarkable.

Conclusion:

Based on this single image, there are no obvious acute abnormalities. However, a definitive diagnosis cannot be made solely on this X-ray. Further evaluation may be necessary to rule out subtle findings or to assess for underlying conditions.

Run the model directly

```
inputs = processor.apply_chat_template(
    messages,
    add_generation_prompt=True,
    tokenize=True,
    return_dict=True,
    return_tensors="pt",
).to(model.device, dtype=torch.bfloat16)

input_len = inputs["input_ids"].shape[-1]

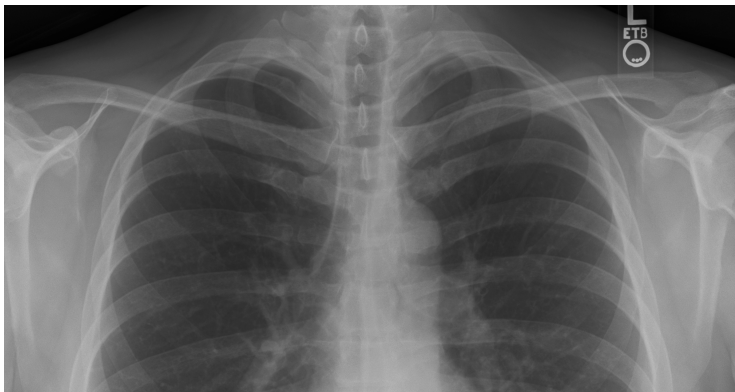
with torch.inference_mode():
    generation = model.generate(**inputs, max_new_tokens=max_new_tokens)
    generation = generation[0][input_len:]

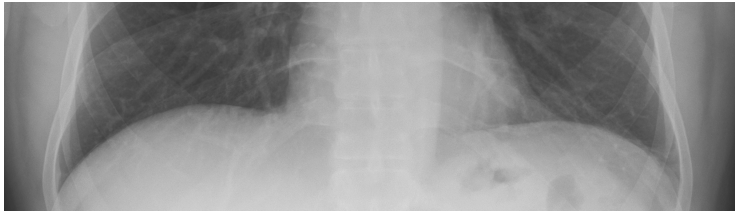
response = processor.decode(generation, skip_special_tokens=True)

display(Markdown(f"---\n\n**[ User ]**\n\n{prompt}"))
display(IPImage(filename=image_filename, height=300))
if is_thinking:
    thought, response = response.split("<unused95>")
    thought = thought.replace("<unused94>thought\n", "")
    display(Markdown(f"---\n\n**[ MedGemma thinking trace ]**\n\n{thought}"))
display(Markdown(f"---\n\n**[ MedGemma ]**\n\n{response}\n\n---"))
```

[User]

Describe this X-ray





[MedGemma]

Okay, I've reviewed the provided chest X-ray. Here's a description of what I see:

Overall Impression:

The X-ray shows a normal adult chest. The lungs appear clear, with no obvious signs of consolidation, effusion, or pneumothorax. The heart size and mediastinal contours are within normal limits.

Specific Findings:

- **Lung Fields:** The lung fields are clear, with no evidence of consolidation, effusion, or pneumothorax.
- **Heart Size:** The heart size appears normal.
- **Mediastinum:** The mediastinum is unremarkable.
- **Ribs:** The ribs appear intact.
- **Clavicles:** The clavicles appear intact.
- **Soft Tissues:** The soft tissues of the chest wall are unremarkable.

In summary, the X-ray is normal.

Important Considerations:

- This is a single view. A full chest X-ray would include both PA (posterior-anterior) and lateral views for a more comprehensive assessment.
- The patient's clinical history is crucial for interpreting the X-ray.
- This interpretation is based solely on the provided image.

Disclaimer: This is a preliminary interpretation based on a single image. A definitive

✓ Run inference on text only

This section demonstrates running inference on text-based tasks.

Specify text prompt and format conversation

```
from IPython.display import Markdown

prompt = "How do you differentiate bacterial from viral infections?"

role_instruction = "You are a helpful assistant."
if is_thinking:
    system_instruction = f"SYSTEM INSTRUCTION: {prompt}"
    max_new_tokens = 1500
else:
    system_instruction = role_instruction
    max_new_tokens = 500
```

prompt

How do you differentiate bacterial from viral infections?

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```

messages = [
    {
        "role": "system",
        "content": [{"type": "text",
    },
    {
        "role": "user",
        "content": [{"type": "text",
    }
]

```

Run model with the `pipeline` API

```

output = pipe(messages, max_new_tokens=max_new_tokens, do_sample=False)
response = output[0]["generated_text"][-1]["content"]

display(Markdown(f"---\n\n**[ User ]**\n\n{prompt}\n\n---"))
if is_thinking:
    thought, response = response.split("<unused95>")
    thought = thought.replace("<unused94>thought\n", "")
    display(Markdown(f"**[ MedGemma thinking trace ]**\n\n{thought}\n'
display(Markdown(f"**[ MedGemma ]**\n\n{response}\n\n---"))

```

[User]

How do you differentiate bacterial from viral pneumonia?

[MedGemma]

Okay, I can explain the key differences between bacterial and viral pneumonia. As a medical assistant, I understand that this is a crucial distinction because the treatment strategies differ significantly. While a definitive diagnosis often requires lab tests, clinical presentation and history can provide strong clues.

Here's a breakdown of how we typically differentiate between the two:

1. Onset & Symptoms:

- **Bacterial Pneumonia:**

- **Onset:** Often **sudden and abrupt**. Patients might feel fine one day and acutely ill the next.
- **Symptoms:**
 - High fever (often > 102°F / 39°C).
 - Shaking chills.
 - Productive cough: Coughing up thick, often colored (yellow, green, rust-colored) sputum.
 - Sharp, stabbing pleuritic chest pain (pain that worsens with deep breaths or coughing).
 - Rapid breathing (tachypnea) and shortness of breath.
 - Fatigue and malaise.
 - May experience sweating.
 - Sometimes confusion (especially in older adults).

- **Viral Pneumonia:**

- **Onset:** Usually **gradual**, often starting with cold-like symptoms that progress over several days.
- **Symptoms:**
 - Lower-grade fever (though it can sometimes be high).
 - Dry, non-productive cough (hacking cough, may produce small amounts of clear or whitish mucus).
 - Muscle aches (myalgia) and headache.
 - Sore throat.
 - General feeling of being unwell (malaise).
 - Shortness of breath, but often less severe initially than bacterial.
 - Chest discomfort may be present but is often less sharp or localized than pleuritic pain.
 - Wheezing can sometimes occur.

2. Patient History & Risk Factors:

Run the model directly

```
processor_or_tokenizer = tokenizer if "text" in model_variant else proc

inputs = processor_or_tokenizer.apply_chat_template(
    messages,
    add_generation_prompt=True,
    tokenize=True,
    return_dict=True,
    return_tensors="pt",
).to(model.device)

input_len = inputs["input_ids"].shape[-1]

with torch.inference_mode():
    generation = model.generate(**inputs, max_new_tokens=max_new_tokens)
    generation = generation[0][input_len:]

response = processor_or_tokenizer.decode(generation, skip_special_tokens=True)

display(Markdown(f"---\n\n**[ User ]**\n\n{prompt}\n\n---"))
if is_thinking:
    thought, response = response.split("<unused95>")
    thought = thought.replace("<unused94>thought\n", "")
    display(Markdown(f"**[ MedGemma thinking trace ]**\n\n{thought}\n\n---"))
display(Markdown(f"**[ MedGemma ]**\n\n{response}\n\n---"))
```

[User]

How do you differentiate bacterial from viral pneumonia?

[MedGemma]

Okay, let's break down how healthcare professionals differentiate between bacterial and viral pneumonia. It's important to remember that I'm an AI and cannot provide medical advice. This information is for educational purposes only. A definitive diagnosis always requires a healthcare professional.

requires a healthcare professional.

Differentiating bacterial and viral pneumonia can be tricky because symptoms often overlap. However, there are key differences in presentation, history, and diagnostic tests that help clinicians distinguish between them.

Here's a breakdown of the key differentiating factors:

1. Clinical Presentation & History:

- **Onset:**

- **Bacterial:** Often sudden and abrupt onset. Patients may feel fine one day and severely ill the next.
- **Viral:** Typically more gradual onset, often starting with cold-like symptoms (runny nose, cough, sore throat) that then progress to pneumonia symptoms.

- **Symptoms:**

- **Bacterial:** High fever (often $>102^{\circ}\text{F}$ / 39°C), shaking chills, productive cough (producing thick, often colored sputum - yellow, green, rust-colored), shortness of breath, chest pain (often sharp, pleuritic - worse with breathing).
- **Viral:** Lower-grade fever (though can be high), dry cough (less likely to produce thick sputum), muscle aches (myalgia), headache, fatigue, sore throat. Shortness of breath can occur but might be less severe initially than in bacterial pneumonia.

- **Patient History:**

- **Bacterial:** Less likely to have preceding upper respiratory infection (URI) symptoms. May have underlying conditions like COPD, heart failure, diabetes, or be immunocompromised.
- **Viral:** Often preceded by URI symptoms. More common in younger children and older adults. Can occur in outbreaks (e.g., influenza season).

2. Physical Examination:

Next steps

Explore the other [notebooks](#) to learn what else you can do with the model.

